INVESTIGATION REPORT

BEYOND THE CRASH

Every Accident Leaves a Story.
Data Reveals the Truth.



NTSB CAROL Aviation Investigation Records

Exploratory Data Analysis | Six Research Cases

Source: https://my.ntsb.gov/aviation

AVIATION ACCIDENT ANALYSIS

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Investigation route

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01 MISSION & DATASET

What the investigation explores

Our objective: identify patterns in documented aviation accidents and compare how frequently events are recorded with how severe their outcomes are.

SOURCE NTSB Aviation Accident Investigation Data (CAROL)

INPUT CAROL_Aviation_Corrected.csv | CSV | 10,000 initial records

PERIOD Accident events from 2 May 2020 to 28 May 2026 (2026 is a partial year)



WHY Flight phases, weather, aircraft attributes, injuries, and investigation findings together enable a compelling, question-led analysis.

10,000	9,995	6,701
Original records	Cleaned records	Analyzed records

NOT ANALYZED 3,294 cleaned records (33.0%) were left out because the analysis keeps only accidents with documented investigation findings. Patterns below describe the 6,701 retained records, not every accident in the source file.

This report examines six research questions. It reports proportions among recorded accidents, not accident probability per flight.

02 DATA PREPARATION

Cleaning, validation and analytical subset

01 Inspect

head(), info(), describe(), missing-value checks, category checks.

02 Clean

Correct selected inconsistencies; harmonize categorical labels; check duplicate IDs.

03 Enrich

Use consistent Make + Model matches for selected aircraft fields and preserve uncertainty elsewhere.

04 Prepare

Split multi-label fields; count each accident once per category; select records with investigation findings.

05 Export

Save both cleaned and analysis CSV files; the original dataset remains unchanged.

QUALITY CHECK

No duplicate investigation IDs remained in the final analysis subset.
Values were not discarded solely for being unusually high.
3,294 cleaned records without documented findings were not analyzed (see Limitations).

03 EXPLORATORY ANALYSIS

Patterns at a glance: distributions, outliers and weather

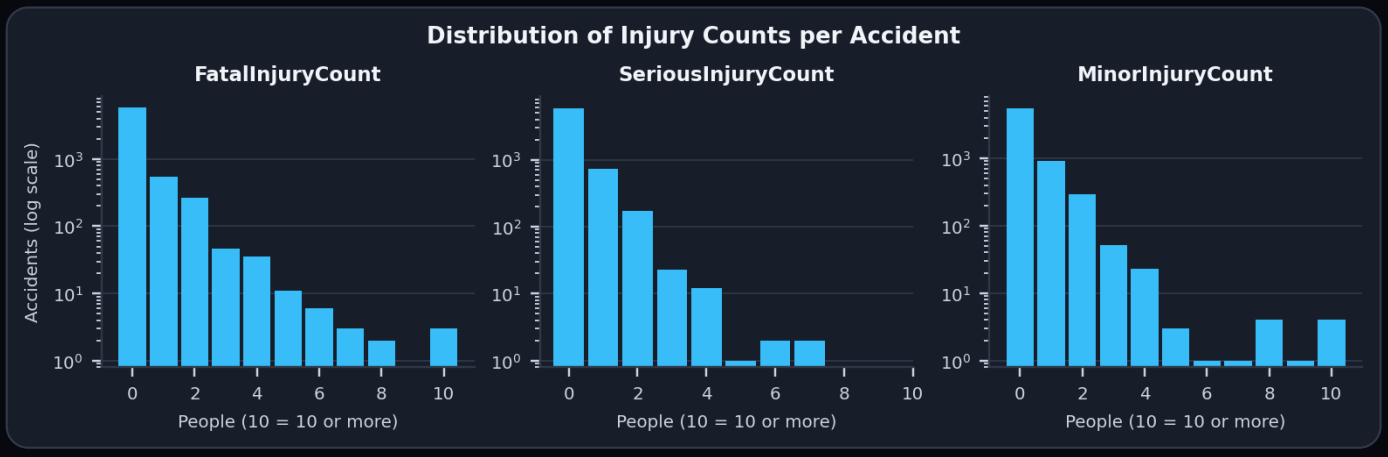


Figure 1. Injury-count histograms; severe right-skew and many zero values. Display choices follow the saved notebook visualization.

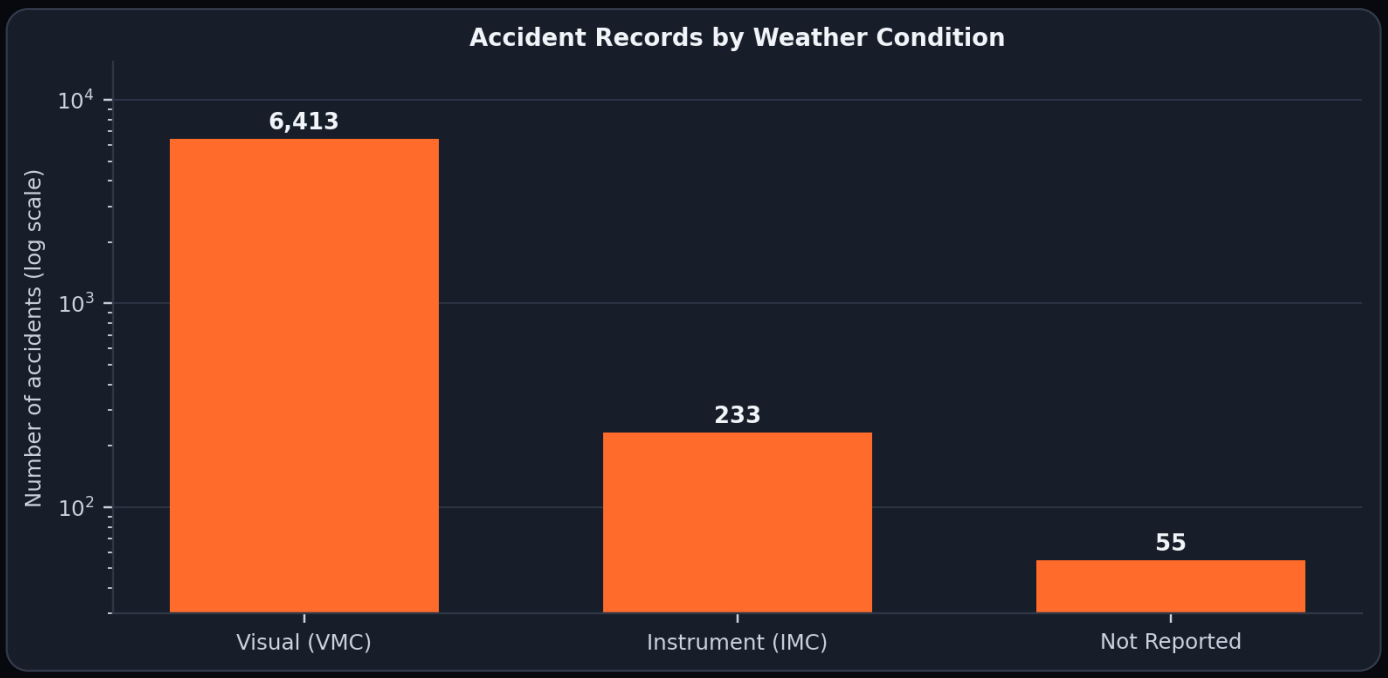


Figure 2. Weather-condition record counts (log scale). These are observed accident counts, not weather-specific flight risks.

04 THE DEADLIEST MOMENT

CASE 01 / Frequency vs fatal outcomes

3,139
Landing records

2.20%
Landing fatal

59.82%
Uncontrolled descent fatal

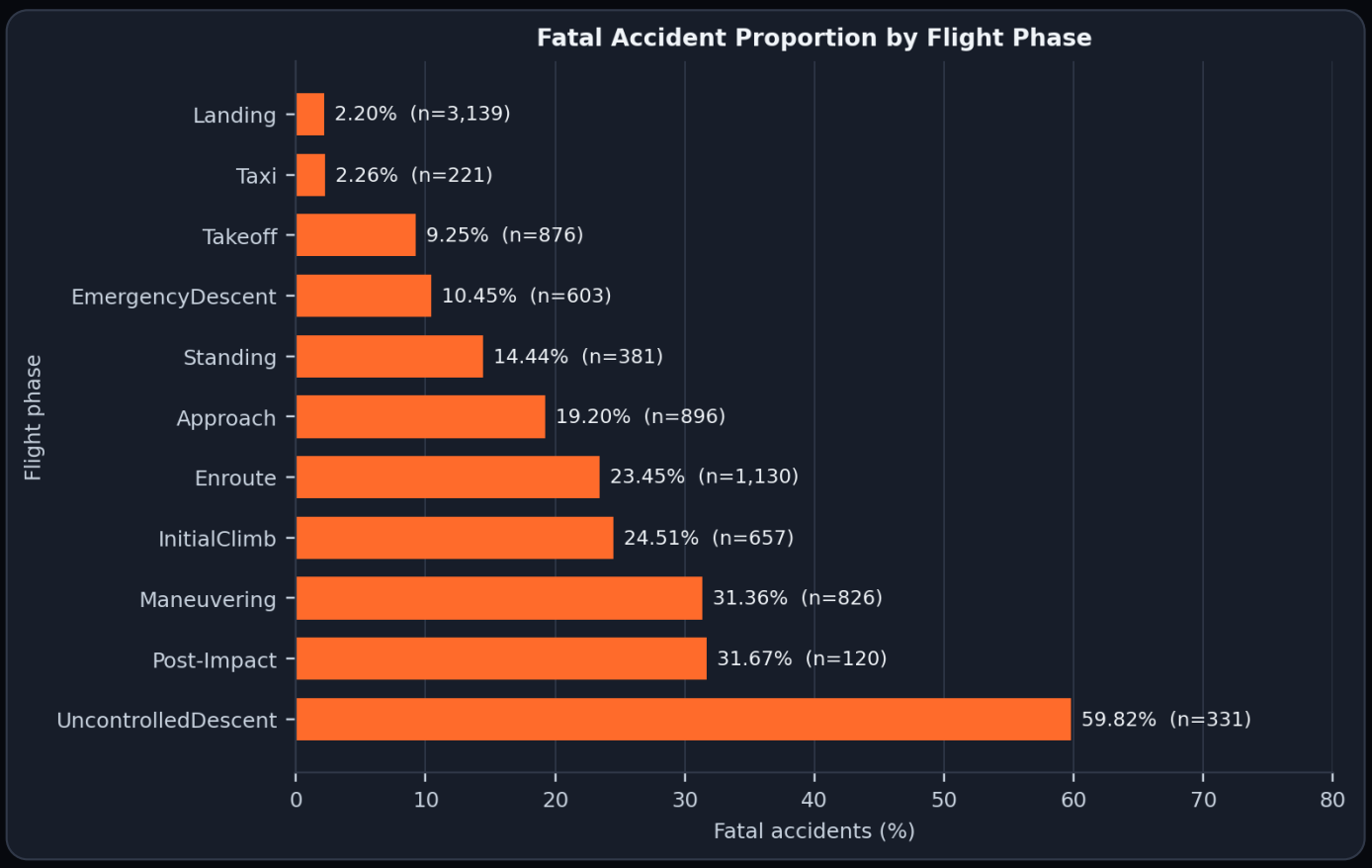


Figure 3. Fatal-accident proportion among incidents associated with each reported flight phase. Phases with fewer than 30 accidents and the 83 records with an unknown phase are not shown.

Interpretation: Landing has the largest count, but uncontrolled descent has the highest fatal proportion: 198 of 331 cases. Multiple phase labels may apply to one accident.

05 THE WEATHER PARADOX

CASE 02 / VMC compared with IMC

11.98%
VMC: 768 / 6,413

59.66%
IMC: 139 / 233

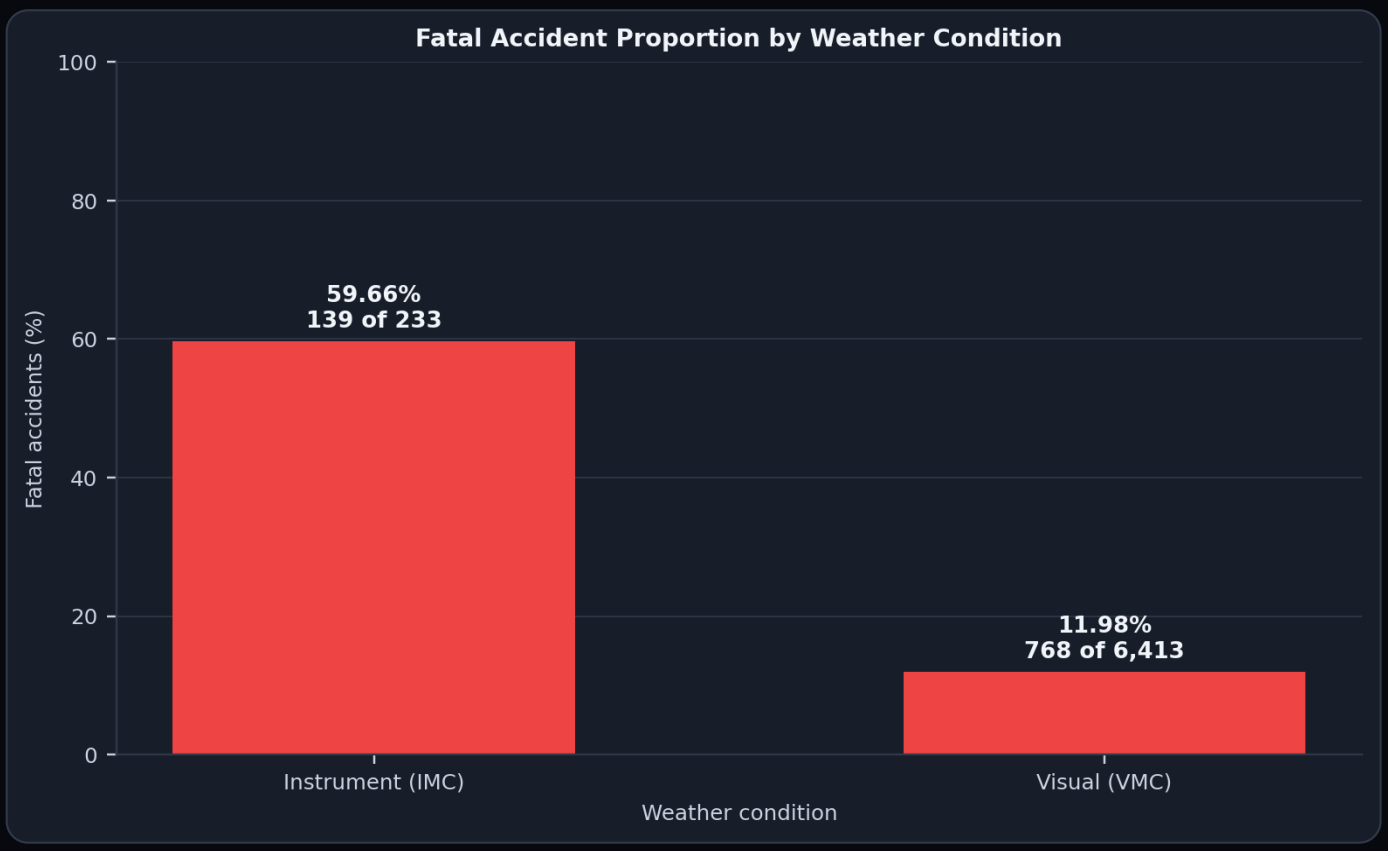


Figure 4. Percent of recorded accidents with at least one fatality in each known weather category. The 55 records with no reported weather are excluded.

Among accidents with documented injury-severity categories, fatal classifications accounted for 29.87% under VMC and 75.14% under IMC. Those values use a smaller, different denominator.

06 FINDINGS & AIRCRAFT

CASES 03-04 / Factors and aircraft characteristics

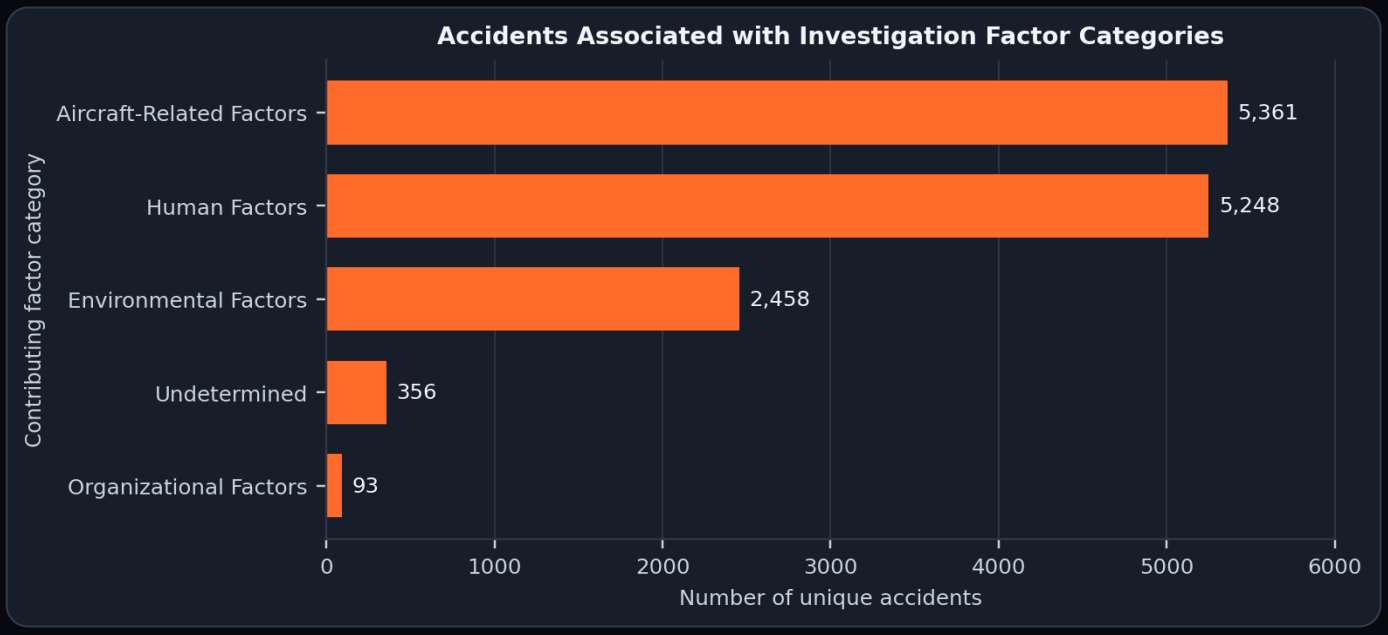


Figure 5. Aircraft-related findings appeared in 5,361 cases; human factors in 5,248. Categories overlap.

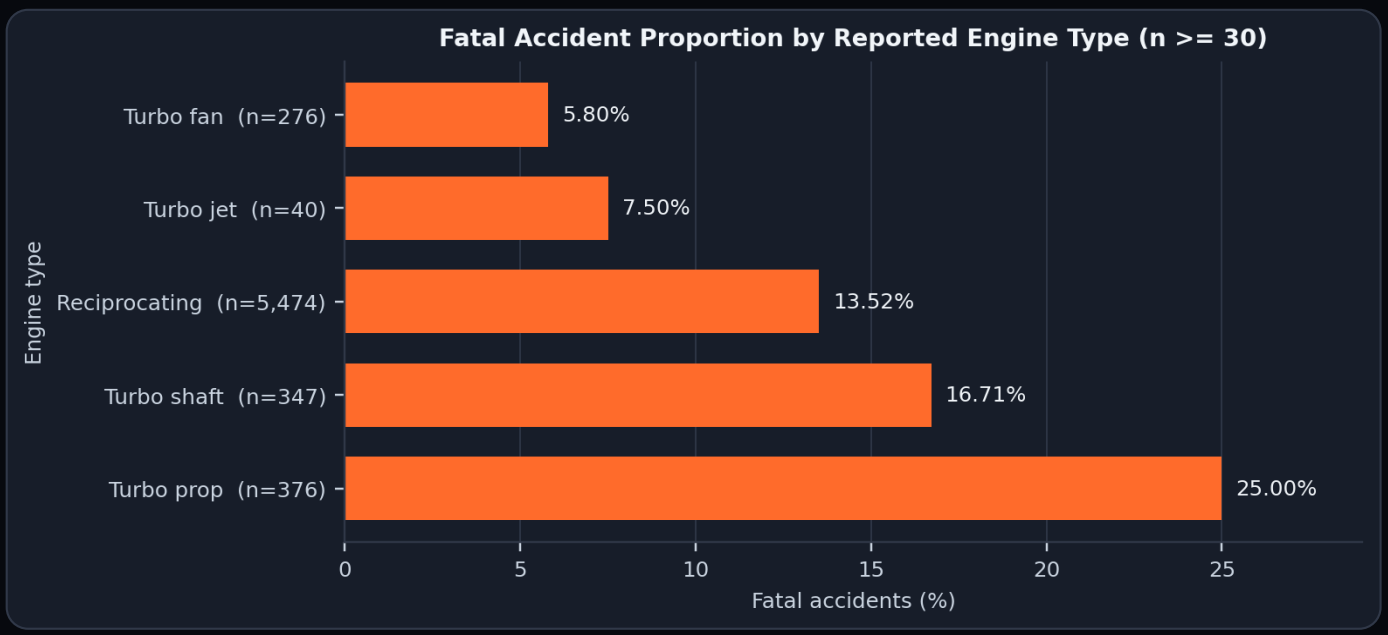


Figure 6. Fatal-accident proportions across engine-type groups with at least 30 accidents; n is shown beside each label. Turbo prop: 25.00%; turbo fan: 5.80%. Reciprocating engines account for most records (n=5,474). The figures do not establish engine-related causality.

07 SURVIVAL & HIDDEN PATTERN

CASES 05-06 / Damage and documented findings

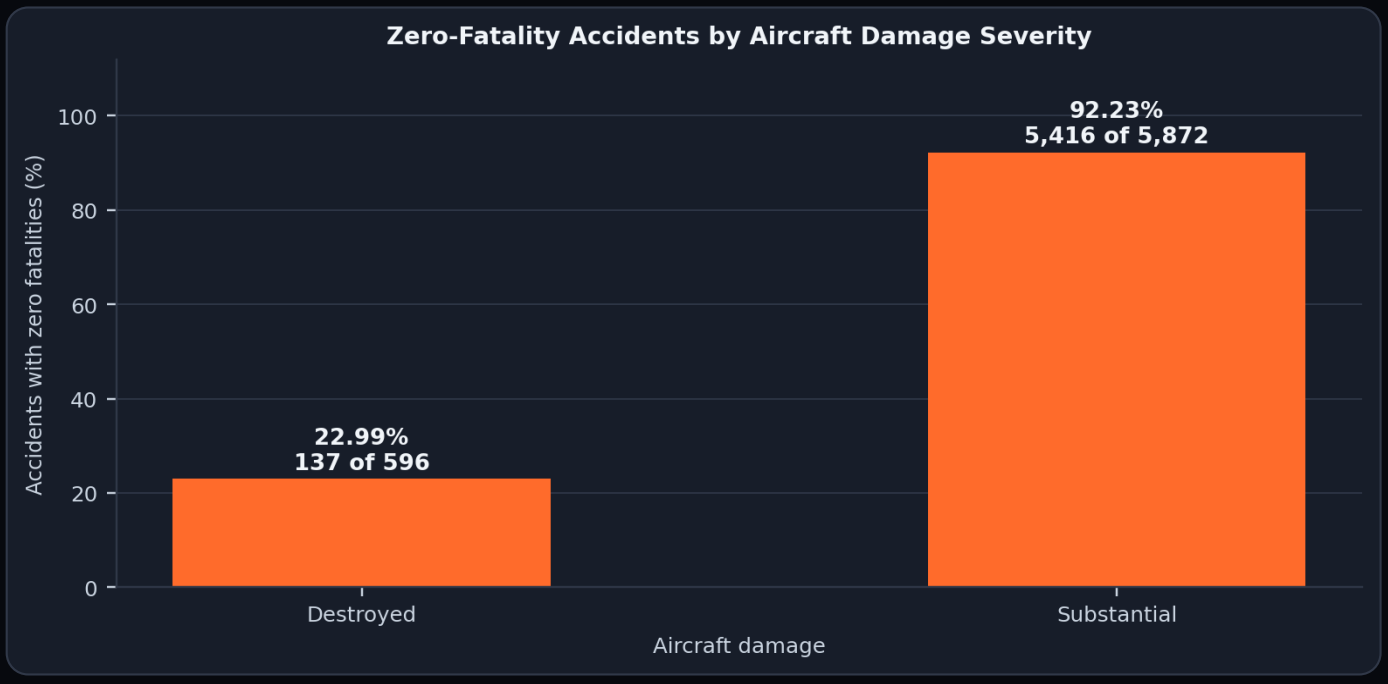


Figure 7. Even when aircraft were classified as destroyed, 137 of 596 accidents (22.99%) recorded zero fatalities; for substantial damage it was 5,416 of 5,872 (92.23%).

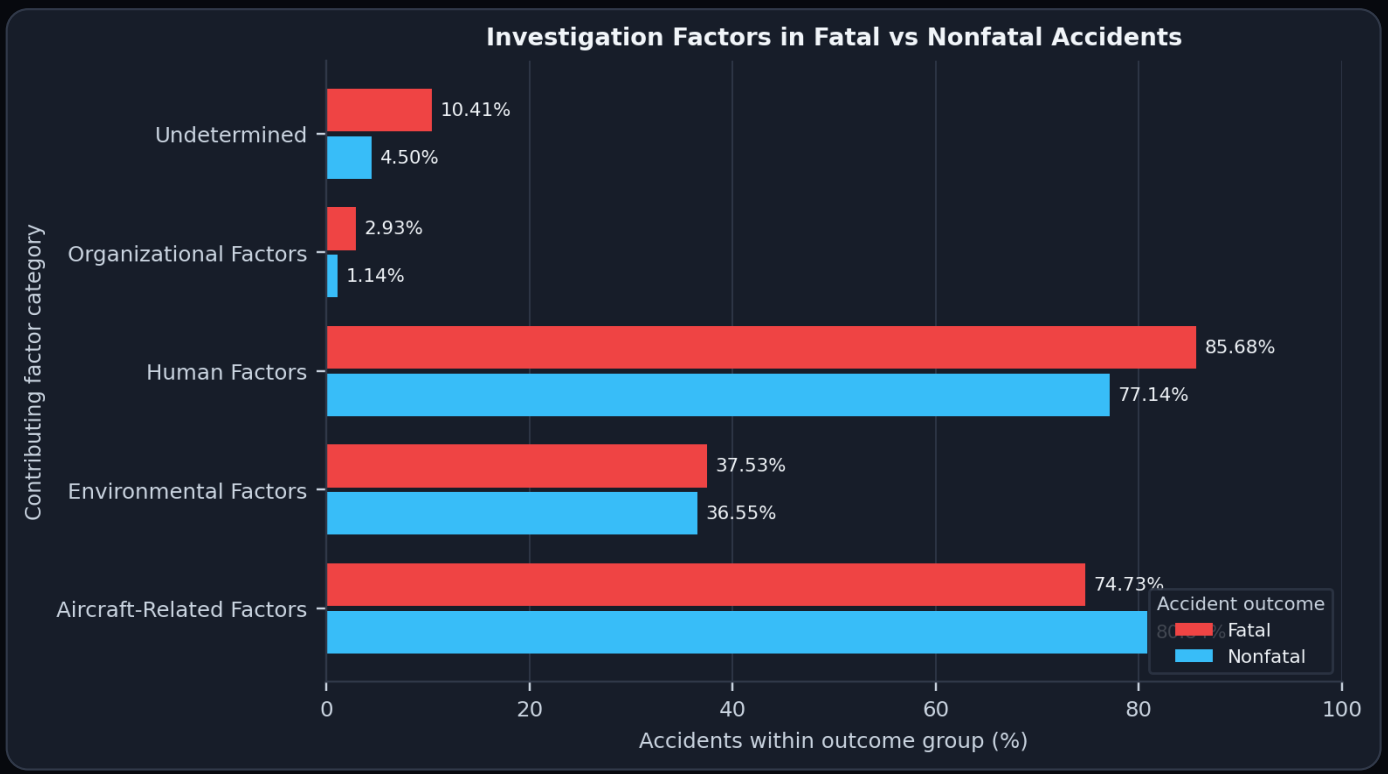


Figure 8. Human findings: 85.68% fatal vs 77.14% nonfatal. Aircraft-related: 74.73% fatal vs 80.84% nonfatal. Fatal n=922; nonfatal n=5,779.

08 METHODS

Analytical workflow and interpretation

DATA LOADING & STRUCTURE

The original CAROL Aviation Corrected CSV was inspected with head(), shape, info(), describe(), and missing-value summaries. The analyzed subset contains 6,701 records; the exported EDA file has 21 columns, including the derived event-year field.

CLEANING & EDA

Cleaning included review of missing information, consistent aircraft make/model lookups, duplicate checks, date preparation, and outlier inspection. The original high injury counts were not deleted solely for being outliers.

VISUAL ANALYSIS

Visual analysis includes histograms, countplots, boxplots, bar charts, line plots, scatter plots and a Spearman correlation heatmap. The notebook also exports cleaned and EDA CSV datasets.

HOW THE FIGURES ARE COUNTED

Multi-label fields (flight phase, findings, aircraft category, engine type) are split on semicolons and each accident is counted once per category. A fatal accident is one with at least one recorded fatality. Rate charts show only groups with at least 30 accidents unless stated otherwise.

09 LIMITATIONS

What these results can and cannot show

DATA COVERAGE

The dataset contained substantial missing information, so columns with little usable content were removed and analyses focused on relevant fields. Events span 2 May 2020 to 28 May 2026; 2026 is a partial year. Recent years have fewer analyzed records (2025: 777; 2026: 234, versus 1,135 in 2024), which may reflect investigations still in progress, so year-to-year trends should not be read as changes in safety.

SELECTION OF RECORDS

Only records meeting the analysis criteria were included: 6,701 of 9,995 cleaned records (3,294 or 33.0% were excluded for lacking documented findings). This can bias the patterns. Some records also required manual review of original investigation reports; unresolved values were not guessed.

SMALL GROUPS AND MISSING LABELS

Some groups are small, so their rates are less stable: IMC accidents (n=233), turbo jet (n=40), uncontrolled descent (n=331). Records with an unreported label are left out of the matching comparison: 83 unknown flight phases, 55 unreported weather conditions, 175 unreported engine types.

DIFFERENT DENOMINATORS

Fatal-accident proportions use all accidents in a group, while injury-severity shares (29.87% VMC, 75.14% IMC) use only accidents with a documented severity level. The two sets of figures are not directly comparable.

INTERPRETATION

Multi-label findings can overlap, and different group sizes affect comparisons. Flight phase, weather, aircraft category and engine type are related to one another, so a difference may reflect more than one factor. Recorded accident proportions are not equivalent to per-flight risk, and none of these associations establish causality.

10 FINAL CONCLUSIONS

What the evidence supports

COUNTS ARE NOT SEVERITY

Landing dominates counts; uncontrolled descent has a much higher fatal proportion among recorded accidents.

CONTEXT MATTERS

IMC accidents show a higher fatal proportion in the analyzed subset; aircraft configurations also differ.

DAMAGE IS NOT DESTINY

Substantial or destroyed aircraft may still be associated with zero recorded fatalities.

FACTOR PATTERNS OVERLAP

Human and aircraft-related investigation findings both appear frequently; neither alone proves a cause.

CONCLUSION

These six analyses show why recorded accident frequency and fatal-outcome proportions should be examined separately. Results describe patterns in the analyzed records, not causes.

REFERENCES

National Transportation Safety Board (NTSB), CAROL Aviation Investigation Data: https://my.ntsb.gov/aviation

NTSB CAROL aviation records, as used in the project notebook.